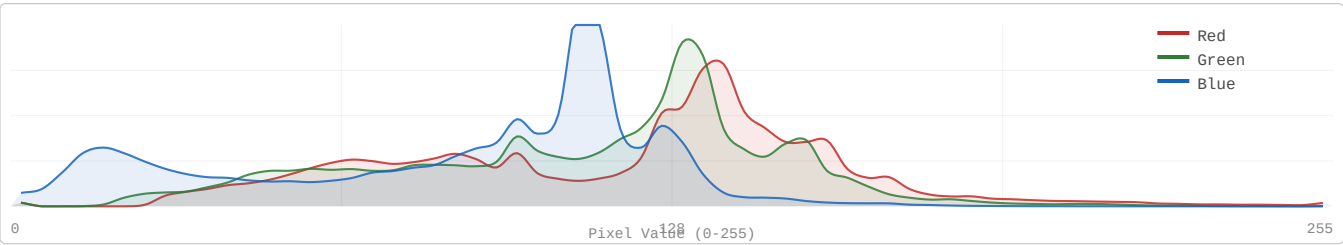


Decorrelation Gap: KODIM07

768x512 | 24-bit RGB | PNG (lossless)

Fruit and Vegetables Display | Weakly one-dimensional (PC1 = 88.77%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

R

G

B

1.0000

0.8330

0.7634

0.8330

1.0000

0.9084

0.7634

0.9084

1.0000

Cb-Cr:

-0.495

(KLT ceiling = 0.000)

Y

Cb

Cr

1.0000

-0.0720

0.0449

-0.0720

1.0000

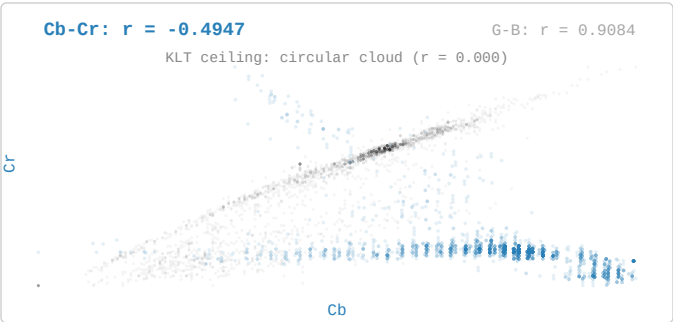
-0.4947

0.0449

-0.4947

1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

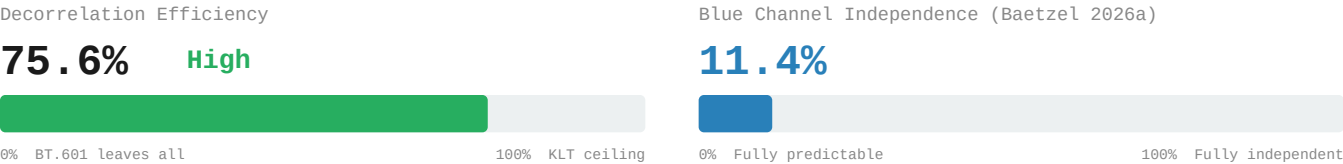
Channel Pair	RGB	YCbCr (BT.601)	Residual r
R-G → Y-Cb	0.832978	-0.072044	0.0720
R-B → Y-Cr	0.763437	0.044902	0.0449
G-B → Cb-Cr	0.908388	-0.494722	0.4947
Avg r	0.834934	0.203889	0.2039

Decorrelation Gap: KODIM07

768x512 | 24-bit RGB | PNG (lossless)

Fruit and Vegetables Display | Weakly one-dimensional (PC1 = 88.77%)

5. Decorrelation Efficiency & Blue Independence



6. 4:2:0 Chroma Subsampling – Information Cost

Channel	PSNR (dB)
R	44.48 dB
G	48.73 dB
B	42.03 dB
Overall	44.29 dB

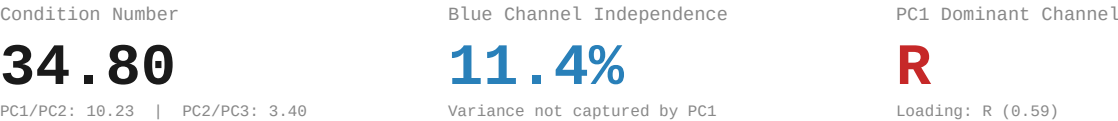
Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)
Eigendecomposition



Eigenvector Loadings			
	R	G	B
PC1	-0.5903	-0.5658	-0.5757
PC2	-0.7823	0.2253	0.5807
PC3	0.1989	-0.7931	0.5756

8. PCA Derived Metrics



9. Redundancy Profile

Classification:	Weakly one-dimensional CN: 34.80 PC1: 88.8%
RGB Correlation:	Avg r = 0.834934 R-G: 0.8330 R-B: 0.7634 G-B: 0.9084
YCbCr Residual:	Avg r = 0.203889 Y-Cb: -0.0720 Y-Cr: 0.0449 Cb-Cr: -0.4947
Efficiency:	75.6% of KLT ceiling Gap: 0.2039 avg r remaining
Blue Channel:	11.4% independent 4:2:0 PSNR: 44.29 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"

[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.

[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.

[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.